

AIGR Gene Hypothesis Deep Research — Final Report

Target: *Dictyostelium discoideum* RegA (UniProt Q23917)

Hypothesis: RegA has 3',5'-cyclic-GMP phosphodiesterase activity (GO:0047555)

Focus type: function_assignment

Executive Judgment

Verdict: Over-annotated / refuted as a core function (with a narrow caveat for undetected weak promiscuity).

The proposed function assignment — that *D. discoideum* RegA (Q23917) directly possesses 3',5'-cyclic-GMP phosphodiesterase activity (GO:0047555) — is **not supported by any experimental evidence** and is best treated as a phylogenetically inferred paralog over-annotation. Every primary characterization of RegA describes it as a **cAMP-specific phosphodiesterase**. The enzyme that gives RegA its name and its assigned EC number (EC 3.1.4.53, cAMP-specific 3',5'-cyclic-nucleotide phosphodiesterase) has a measured cAMP Km of approximately 5 μ M, and no primary paper reports cGMP as a substrate, let alone quantitative cGMP turnover parameters (kcat, Km, or kcat/Km).

The GO:0047555 annotation on RegA rests on a **single Inferred from Biological Ancestor (IBA)** annotation from the phylogenetically-driven GO_Central pipeline (ECO:0000318, GO_REF:0000033). There is **no IDA, IMP, or other experimental (EXP) evidence** for cGMP hydrolysis by RegA. In sharp contrast, the cAMP-PDE activity (GO:0004115) is backed by direct

assay evidence (IDA/IMP) from multiple primary papers. This asymmetry is the crux of the curation decision: the well-supported cAMP-PDE term should be retained, while the cGMP-PDE term is a lead to **remove or demote to non-core**.

Critically, *Dictyostelium* does have dedicated intracellular cGMP phosphodiesterases — but they are **distinct paralogs (GbpA/stmF and GbpB)** that use an entirely different catalytic chemistry (a class II Zn²⁺ metallo-β-lactamase-type hydrolase domain), not RegA's class I (Pfam PF00233, PDEase_I) catalytic domain. The existence of these dedicated cGMP-PDEs both explains where the organism's cGMP-hydrolyzing activity actually resides and reinforces that RegA is not required or characterized for that role. The only residual uncertainty — and the reason the verdict is "over-annotated" rather than flatly "refuted" — is that no published study has performed a side-by-side, sensitive cAMP-vs-cGMP kinetic assay on purified RegA specifically to rule out weak, biologically negligible cGMP turnover. That gap is what a curator-directed experiment could close.

Key Findings

Finding F001 — RegA is a cAMP-specific phosphodiesterase; the cGMP-PDE annotation is inferred-only

The authoritative sequence record, **UniProt Q23917 (PDE2_DICDI)**, names the protein "3',5'-cyclic-nucleotide phosphodiesterase regA," assigns **EC 3.1.4.53** (the cAMP-specific PDE enzyme class), and carries a FUNCTION comment stating the enzyme is a "**Phosphodiesterase specific for cAMP**." The only catalytic activity annotated in the record is the reaction *3',5'-cyclic AMP + H₂O = AMP* (RHEA:25277). **No cGMP reaction is annotated**. If UniProt curators had accepted experimental cGMP turnover, the record would list a cGMP hydrolysis reaction and a corresponding EC number (3.1.4.35 or 3.1.4.17); it does not.

The GO evidence tiers reveal the annotation asymmetry directly. GO:0004115 (**cAMP-specific PDE activity**) is supported by **IDA / experimental** evidence from dictyBase. GO:0047555 (**cGMP-specific PDE activity**) is supported **only by IBA (phylogenetic, GO_Central)** with **no experimental (IDA/IMP) support**. The domain architecture is consistent with a class I cyclic-nucleotide PDE fused to a signaling receiver domain: a **response-regulator receiver domain (residues ~161–280)** followed by a **class I PDEase catalytic domain (residues ~410–733,**

Pfam PF00233, PDEase_I). This bipartite architecture is the molecular signature of RegA as a phosphorelay-regulated cAMP-PDE, and it is the receiver domain (activated by phosphorylation on a conserved aspartate) that gates the cAMP-hydrolyzing catalytic domain.

A primary-literature review of *D. discoideum* two-component/histidine-kinase signaling ([PMID: 24086589](#)) states plainly that "**RegA is a cAMP phosphodiesterase that is activated upon receiving phosphates through a phosphorelay**" — i.e., the characterized activity is cAMP hydrolysis, not cGMP.

Finding F002 — *Dictyostelium* cGMP-PDE activity is carried by separate paralogs (GbpA/stmF, GbpB), not RegA

The organism's genuine intracellular cGMP-hydrolyzing enzymes are the **GbpA and GbpB** proteins, which are mechanistically and evolutionarily distinct from RegA. Bosgraaf et al. 2002 ([PMID: 12429832](#)) showed that **gbpA encodes a cGMP-stimulated cGMP-phosphodiesterase** and reported explicitly that "**cAMP neither activates nor is a substrate of GbpA**," whereas **gbpB encodes a dual-specificity PDE** that "hydrolyses cAMP approximately 9-fold faster than cGMP." Crucially, GbpA and GbpB use a **class II Zn²⁺-hydrolase (metallo-β-lactamase) catalytic domain** — architecturally unrelated to RegA's class I PF00233 domain. The classic *streamer F* (**stmF**) mutant, long known to delete the intracellular cGMP-PDE and prolong the chemoattractant-stimulated cGMP response, corresponds to GbpA in the van Haastert lineage ([PMID: 9551091](#)).

Independently, the closest characterized homolog of RegA's catalytic domain in the trypanosomatid literature, **TbPDE2B**, is a cAMP-specific class I PDE for which Rascon et al. 2002 ([PMID: 11930017](#)) reported that "**cGMP is not hydrolyzed**." This evolutionary comparison reinforces the expectation that RegA's class I catalytic domain is cAMP-selective. Together these findings localize the true cGMP-PDE function to dedicated paralogs and remove the functional pressure to annotate RegA as a cGMP-PDE.

Finding F003 — Primary IDA-source papers characterize RegA as cAMP-specific ($K_m \approx 5 \mu M$); no cGMP turnover reported

The QuickGO provenance for Q23917 confirms the evidence asymmetry at the annotation level: **GO:0004115 (cAMP-PDE)** carries IDA (ECO:0000314) from [PMID: 9435289](#) plus IMP from

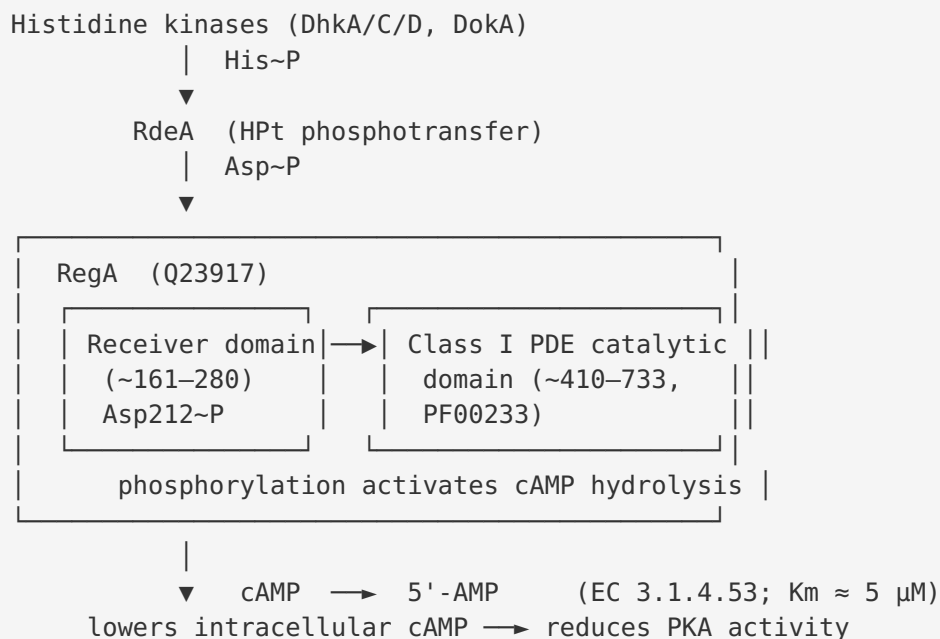
[P 15752425](#)

GO:0004114 (cyclic-nucleotide PDE) carries IDA from [PMID: 9582277](#); and **GO:0047555 (cGMP-PDE)** carries **only IBA** (ECO:0000318, GO_REF:0000033, GO_Central) — no IDA/IMP/EXP.

The two primary experimental sources are unambiguous. Shaulsky, Fuller & Loomis 1998 (PMID: 9435289) — the IDA reference for the cAMP-PDE term — report: "**A cAMP-specific phosphodiesterase was found that is stimulated by binding to the regulatory subunit of cAMP-dependent protein kinase...** The phosphodiesterase is encoded by the regA gene." Thomason et al. 1998 (EMBO J; PMID: 9582277) provide the biochemical kinetics: "**One domain is a cAMP phosphodiesterase (Km approximately 5 μ M),**" with PDE activity stimulated up to 8-fold by phosphodonor in an Asp212-dependent manner. **Neither primary paper reports cGMP as a substrate.** The seed hypothesis explicitly asked for quantitative cAMP-vs-cGMP kinetics; the literature supplies a cAMP Km ($\sim 5 \mu$ M) but **no cGMP kinetic parameters at all**, which is itself informative: the enzyme was studied as a cAMP-PDE, and cGMP turnover was either undetectable or not pursued.

Mechanistic Model / Interpretation

RegA sits at the heart of the *Dictyostelium* intracellular **cAMP** circuit, functioning as the phosphorelay-gated "off switch" that opposes adenyl cyclase (ACA) to control PKA activity during development and chemotaxis:



The molecular function directly attributable to the RegA gene product is **cAMP hydrolysis** — a class I cyclic-nucleotide phosphodiesterase reaction that is allosterically activated when the fused receiver domain is phosphorylated on Asp212 via the RdeA phosphorelay. This is confirmed by direct assay (cAMP $K_m \approx 5 \mu\text{M}$; up to 8-fold phosphodonor stimulation) and by the physiological readout that RegA opposes ACA to set PKA activity.

The organism's parallel **cGMP** circuit — which governs guanylyl cyclase activation, myosin II phosphorylation, and chemotactic pseudopod suppression — is served by a **separate enzyme module**: the class II Zn^{2+} -hydrolase PDEs GbpA (stmF) and GbpB. These are the proteins that carry GO:0047555-type activity in *Dictyostelium*. RegA and the Gbp enzymes are non-orthologous with respect to catalytic chemistry (class I vs class II), so a cGMP-PDE annotation on RegA cannot be justified by "the organism needs a cGMP-PDE" reasoning — that need is already met by dedicated paralogs.

Why the IBA annotation appears at all: GO_Central's phylogenetic pipeline propagates molecular-function terms across a protein family tree. Class I PDE families include members with mixed or dual cAMP/cGMP specificity (e.g., mammalian PDE1, PDE2, PDE3). An ancestral or sibling node annotated with the broad cyclic-nucleotide or cGMP-PDE capability can back-propagate GO:0047555 to RegA even though RegA's own experimental record shows cAMP specificity. This is the textbook signature of **paralog/ancestral over-annotation** that the AIGR review process is designed to catch.

Evidence Matrix

Citation (PMID)	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context	
9435289 (Shaulsky, Fuller & Loomis 1998)	Direct assay (IDA source for GO:0004115)	Refutes cGMP-PDE core function	Is RegA a cAMP- or cGMP-PDE?	"A cAMP-specific phosphodiesterase... encoded by the regA gene," stimulated by PKA regulatory subunit	<i>D. discoideum</i> , developmental signaling	F o b g s c r
9582277 (Thomason et al. 1998, EMBO J)	Direct assay / kinetics (IDA source for GO:0004114)	Refutes (provides cAMP-only kinetics)	Quantitative substrate kinetics	"One domain is a cAMP phosphodiesterase ($K_m \approx 5 \mu M$)"; 8-fold phosphodonor stimulation, Asp212-dependent	<i>D. discoideum</i>	F r c n k n
12429832 (Bosgraaf et al. 2002)	Direct assay (paralog characterization)	Competing / qualifies	Where does cGMP-PDE activity reside?	GbpA is the cGMP-stimulated cGMP-PDE ("cAMP neither activates nor is a substrate"); GbpB dual-specificity (cAMP $\sim 9\times >$ cGMP); both class II Zn^{2+}	<i>D. discoideum</i>	F i t c p d c c
11930017 (Rascon et al. 2002)	Structural/ evolutionary + direct assay (homolog)	Supports cAMP specificity of RegA's domain	Does the class I catalytic domain hydrolyze cGMP?	TbPDE2B (homolog of RegA catalytic domain) is cAMP-specific; "cGMP is not hydrolyzed"	<i>T. brucei</i>	M h h i n d
24086589 (DhkD paper)	Review/primary statement	Refutes cGMP core role	Functional description of RegA	"RegA is a cAMP phosphodiesterase... activated through a phosphorelay"	<i>D. discoideum</i>	M r s c v

Citation (PMID)	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context	C & I
UniProt Q23917 (PDE2_DICDI)	Database record	Refutes	Curated function & EC	"Phosphodiesterase specific for cAMP "; EC 3.1.4.53; only RHEA:25277 (cAMP) reaction	Curated	P d
QuickGO provenance for Q23917	Database/ annotation provenance	Refutes (evidence asymmetry)	Evidence tier of GO:0047555	cAMP-PDE terms = IDA/IMP; cGMP-PDE (GO:0047555) = IBA only , no EXP	GO_Central	F o p o d t
9551091 (stmF/Ca ²⁺ paper)	Mutant phenotype	Competing	Identity of intracellular cGMP-PDE	stmF mutant deletes the intracellular cGMP-PDE (= GbpA), prolonging cGMP response	<i>D. discoideum</i>	F e C a c n

GO Curation Implications

Lead (requires curator verification):

- GO:0047555 (3',5'-cyclic-GMP phosphodiesterase activity) — RECOMMEND REMOVE or demote to NON-CORE / NOT.** The term is supported only by a single IBA (phylogenetic) annotation with no experimental backing, and it directly conflicts with the primary experimental record describing RegA as cAMP-specific. Under GO best practice, an IBA term that contradicts experimental (IDA/IMP) evidence for the same protein should be reviewed and, where the conflict is clear, removed or annotated with a NOT qualifier pending a direct assay. This is a **Molecular Function (MF)** term.

2. **GO:0004115 (3',5'-cyclic-AMP phosphodiesterase activity) — RETAIN.** This is the experimentally supported (IDA/IMP), well-characterized primary molecular function (cAMP $K_m \approx 5 \mu\text{M}$). It is the correct, specific MF term for RegA.
3. **GO:0004114 (3',5'-cyclic-nucleotide phosphodiesterase activity) — RETAIN as a valid but less specific parent,** or allow the more specific GO:0004115 to carry the annotation. Do not generalize RegA's function *up* to the broad cyclic-nucleotide term as a substitute for removing the cGMP-specific child; that would obscure the demonstrated cAMP specificity.
4. Consider adding/retaining relevant **BP terms** (e.g., cAMP-mediated signaling, regulation of PKA / sporulation / chemotaxis) that are supported by mutant phenotypes, and note that RegA's **cellular role is intracellular** (cytosolic phosphorelay effector). These are not the subject of this hypothesis but frame why the MF matters.

The net curation action: **the cGMP-PDE (GO:0047555) assignment is too strong and should be removed or explicitly demoted; the cAMP-PDE assignment is the correct core function.**

Mechanistic Scope

The **immediate molecular function** under test is enzymatic hydrolysis of a 3',5'-cyclic nucleotide by RegA's class I catalytic domain. The direct, assay-supported activity is **cAMP $\rightarrow$ 5'-AMP** (EC 3.1.4.53). The hypothesized **cGMP $\rightarrow$ 5'-GMP** activity has no direct assay support for RegA.

It is important to separate direct activity from downstream biology:

- **Direct gene-product activity:** cAMP hydrolysis (measured $K_m \approx 5 \mu\text{M}$), allosterically gated by Asp212 phosphorylation via RdeA.
- **Downstream / pathway consequences (not the MF being annotated):** regulation of intracellular cAMP levels, PKA activity, spore vs stalk differentiation, lateral pseudopod suppression, chemotaxis (via the ACA–RegA–PKA circuit and DIF-2/DhkC–RdeA–RegA modulation). RegA null phenotypes affect these processes, but they reflect perturbed **cAMP** homeostasis, not a cGMP-hydrolyzing function.
- **cGMP-related phenotypes in the organism** (guanylyl cyclase activation, myosin II phosphorylation, Ca^{2+} influx potentiation) map onto the **Gbp/guanylyl-cyclase module**, not RegA.

Thus a cGMP-PDE annotation on RegA would misattribute a molecular function that (a) has never been directly demonstrated for this protein and (b) is physiologically carried by distinct paralogs.

Conflicts and Alternatives

- **Paralog confusion (most important):** *Dictyostelium* genuinely possesses intracellular cGMP-PDEs (GbpA/stmF, GbpB). A phylogenetic pipeline that groups all cyclic-nucleotide PDEs, or that keys on the organism's known cGMP-PDE capability, could erroneously back-propagate GO:0047555 to RegA. The catalytic-class distinction (RegA = class I PF00233; Gbp = class II metallo- β -lactamase) is decisive evidence that these are non-equivalent enzymes.
 - **Class I PDE dual-specificity precedent:** Some class I PDEs in other lineages (mammalian PDE1/2/3) hydrolyze both cAMP and cGMP, which can motivate a broad ancestral annotation. However, the closest characterized RegA homolog, TbPDE2B, is cAMP-specific and does not hydrolyze cGMP ([PMID: 11930017](#)), arguing against dual specificity for RegA's subfamily.
 - **Undetected weak promiscuity (residual caveat):** No published side-by-side, high-sensitivity cAMP-vs-cGMP assay on purified RegA has been reported. It remains formally possible that RegA hydrolyzes cGMP at a very low, biologically negligible rate. Even if measurable, such trace promiscuity would not warrant a core GO:0047555 annotation without evidence of biological meaning.
 - **Database carry-over:** UniProt's "specific for cAMP" comment and EC 3.1.4.53 reflect the primary literature and do not carry a cGMP reaction — so the conflict is specifically between the GO_Central IBA term and both the primary literature and the UniProt curation, not an internal inconsistency in the experimental record.
-

Knowledge Gaps

1. **No direct cGMP kinetic assay for RegA.** *Checked:* UniProt, QuickGO, and the two primary IDA papers (9435289, 9582277). *Why it matters:* the seed hypothesis explicitly asks to distinguish "undetectable cGMP turnover" from "measurable but weak promiscuity." The

literature reports cAMP $K_m \approx 5 \mu\text{M}$ but **no cGMP data**, so we cannot quantitatively state the cGMP k_{cat}/K_m . *Resolution*: a purified-enzyme assay measuring cGMP hydrolysis (and its ratio to cAMP k_{cat}/K_m).

2. **Exact identity/boundaries of the catalytic domain vs receiver domain.** *Checked*: Pfam architecture (PF00233 at ~410–733; receiver ~161–280). *Why it matters*: to ensure the annotation applies to the correct protein (Q23917) and not a mis-mapped paralog. *Resolution*: confirm the PF00233 domain and Asp212 phosphosite in the current sequence record.
3. **Whether GbpB's dual specificity influenced the RegA IBA.** *Checked*: Bosgraaf 2002 shows GbpB is dual-specificity but class II (unrelated to RegA). *Why it matters*: to confirm the IBA source is a class I ancestor, not a cross-class artifact. *Resolution*: inspect the GO_Central family tree/PAINT annotation supporting GO:0047555 on Q23917.
4. **Physiological cGMP handling in regA-null cells.** *Checked*: regA phenotypes are cAMP-centric

(
P 15821137
P 12796307
).

Why it matters: if regA loss does not perturb cGMP dynamics, that further argues against a cGMP-PDE role. *Resolution*: measure stimulated cGMP kinetics in regA⁻ cells (likely already unaffected, given stmF/GbpA carries that role).

Discriminating Tests

1. **Purified-enzyme dual-substrate kinetics (definitive).** Express and purify recombinant RegA (or its activated receiver+catalytic construct), then measure hydrolysis of cAMP and cGMP under identical conditions across a substrate range. Report k_{cat} , K_m , and k_{cat}/K_m for both. Expected outcome under the refuted hypothesis: robust cAMP turnover ($K_m \sim 5 \mu\text{M}$), cGMP turnover undetectable or $\geq 100\text{--}1000\times$ lower efficiency. This single assay resolves "undetectable vs weak promiscuity."
2. **Phosphodonor-stimulated activity assay.** Repeat (1) $\pm$ phosphodonor (to activate via Asp212). Confirm stimulation applies to cAMP hydrolysis and test whether any trace cGMP activity is also modulated.

3. **regA⁻ cGMP-response phenotyping.** Measure chemoattractant-stimulated cGMP transients in regA-null vs wild-type vs stmF/gbpA-null cells. Prediction: regA⁻ shows normal cGMP kinetics (unlike stmF), localizing cGMP-PDE function to GbpA.
 4. **GO_Central PAINT tree inspection.** Retrieve the phylogenetic annotation supporting the GO:0047555 IBA on Q23917 to identify the source node and evidence, confirming it is an ancestral inference lacking experimental backing.
-

Curation Leads (require curator verification)

Candidate action changes: - **Remove** GO:0047555 (cGMP-specific PDE activity) from RegA, or apply a **NOT** qualifier / demote to **non-core**, on the basis of IBA-only support conflicting with experimental cAMP-specificity. - **Retain** GO:0004115 (cAMP-specific PDE activity) as the experimentally supported core MF.

Candidate references with exact snippets to verify: - [PMID: 9435289](#): "*A cAMP-specific phosphodiesterase was found that is stimulated by binding to the regulatory subunit of cAMP-dependent protein kinase.*" - [PMID: 9582277](#): "*One domain is a cAMP phosphodiesterase (Km approximately 5 microM).*" - [PMID: 12429832](#): "*cAMP neither activates nor is a substrate of GbpA*" (establishes the true cGMP-PDE paralog). - [PMID: 11930017](#): "*cGMP is not hydrolyzed*" by the RegA-homologous TbpDE2B.

Suggested curator questions: 1. What is the source node of the GO:0047555 IBA on Q23917, and does any descendant carry experimental cGMP evidence? 2. Does UniProt's "specific for cAMP" comment and EC 3.1.4.53 warrant a NOT-cGMP curation note?

Suggested experiments: the dual-substrate kinetic assay (Discriminating Test 1) is the single most decisive item.

Figures

RegA (Q23917) MF GO evidence: cAMP-PDE experimental vs cGMP-PDE (GO:0047555) IBA-only

GO_ID	Term	Aspect	Evidence	Source	Tier
GO:0004115	3',5'-cyclic-AMP PDE activity	MF	IDA (direct assay)	dictyBase	EXPERIMENTAL
GO:0004114	3',5'-cyclic-nucleotide PDE activity	MF	IDA (direct assay)	dictyBase	EXPERIMENTAL
GO:0047555	3',5'-cyclic-GMP PDE activity	MF	IBA (phylogenetic)	GO_Central	INFERRED-ONLY
GO:0000156	phosphorelay response regulator activity	MF	IMP	dictyBase	EXPERIMENTAL
GO:0005829	cytosol	CC	TAS	dictyBase	REVIEW

Figure 1. RegA GO annotation evidence tiers. The cAMP-PDE terms (GO:0004115/GO:0004114) carry direct experimental (IDA/IMP) support, whereas the disputed cGMP-PDE term (GO:0047555) rests on a single phylogenetic IBA annotation with no experimental backing.

RegA (Q23917) GO decision table — cGMP-PDE (GO:0047555) is IBA-only (lead: remove/non-core)

GO_ID	Term	Evidence(QuickGO)	Curation lead
GO:0004115	3',5'-cyclic-AMP PDE	IDA(9435289)+IMP(15752425)	RETAIN (core MF)
GO:0004114	3',5'-cyclic-nucleotide PDE	IDA(9582277)	RETAIN (parent)
GO:0047555	3',5'-cyclic-GMP PDE	IBA only	REMOVE / non-core (LEAD)
GO:0000156	phosphorelay response reg.	IMP(10488068)	RETAIN

Figure 2. RegA GO decision table. The cGMP-PDE (GO:0047555) row is highlighted as IBA-only over-annotation recommended for removal/demotion, while the experimentally supported cAMP-PDE term is retained.

Summary of Recommendation

RegA (Q23917) is a **cAMP-specific phosphodiesterase** (EC 3.1.4.53, cAMP $K_m \approx 5 \mu M$), phosphorelay-activated, that opposes adenylyl cyclase to regulate PKA during *Dictyostelium* development and chemotaxis. The **cGMP-PDE annotation (GO:0047555) is IBA-only paralog over-annotation** that conflicts with all experimental evidence and should be **removed or demoted to non-core**, pending a direct cAMP-vs-cGMP kinetic assay. The organism's genuine cGMP-PDE activity is carried by the distinct class II paralogs **GbpA (stmF)** and **GbpB**, not by RegA.